

APPLICATION FOR ETHICS APPROVAL

No ethics-relevant action should be taken until this application has been approved.

Please

1. first read the [Guidelines](#) to determine whether approval is required,
2. fill out this form following the instructions in the [Guidelines](#),
3. add any additional information in separate files, and
4. send the complete application to erp-submissions@uni.lu.

SECTION 1: PROJECT

Title:

Acronym:

Applicant(s):

Researcher/Student	Name:	Jules Wax
	Position at UL:	Doctoral Candidate
	Faculty / IC:	FSTM
	Research Unit:	SnT / TRUX
	Telephone no:	+352 621 784 440
	Email:	jules.wax@uni.lu
Principal Investigator ¹	Name:	Jacques Klein
	Position at UL:	Professor
	Faculty / IC:	FSTM
	Research Unit:	SnT / TRUX
	Telephone no:	
	Email:	jacques.klein@uni.lu

*Please specify in case you are not employed at UL.
Please expand if more than 2 persons are involved.*

¹ if different from researcher; Supervisor if the researcher is a PhD student

Funding:

Source:

Is ethical approval required before the funds can be obtained?

☐ Yes

☒ No

Ethical approval is being requested for:

☒ a single
study

☐ multiple
studies

Please note that enough detail needs to be provided for each study.

Has ethics approval already been obtained from another ethics committee for this project?

☐ Yes

☒ No

Has ethics approval already been obtained from the ERP for a similar research design?

☐ Yes

☒ No

In case ethics approval has already been obtained, please provide references.
(see points below)

- ...[the reference number of the ERP](#).
- ...[from another ethics committee, please join a copy of the approval letter](#).
- ...

Please list any **research area-specific ethics guidelines** (with a link to the relevant document) you have used in the planning of your research (e.g. from a professional society).

- ...
- ...

SECTION 2: SHORT SUMMARY OF THE PROJECT

(including research design and methodology)

Research has found that there are two different trust measures: Cognitive Trust and Emotional Trust. The former refers to trust triggered by model understanding and/or explanation, while the latter refers to trust in whether the model is understood or has been explained. The most assessed measure is Cognitive Trust. The purpose of this project is therefore to evaluate the level of trust in AI systems by appealing to emotions.

The project will involve assessment of participants through a general trust scale (TXAI) and an emotional trust scale in AI.

Participants will be asked to fill in a short survey assessing their general and emotional trust regarding AI. They will then be confronted with an artistic visualization of the AI process from input to output. Following this, they will be asked the same survey questions as before. The purpose will be to see whether the artistic depiction and explanation enhanced, hindered, or did not affect their trust in AI (including emotional trust).

References

Perrig, Sebastian AC, Nicolas Scharowski, and Florian Brühlmann. "Trust issues with trust scales: examining the psychometric quality of trust measures in the context of AI." *Extended abstracts of the 2023 CHI Conference on human factors in computing systems*. 2023.
<https://doi.org/10.1145/3544549.3585808>

Shang, Ruoxi, Gary Hsieh, and Chirag Shah. "Trusting your ai agent emotionally and cognitively: Development and validation of a semantic differential scale for ai trust." *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. Vol. 7. 2024.
<https://doi.org/10.1609/aies.v7i1.31728>

SECTION 3: PROPOSED START DATE AND DURATION OF ETHICS-RELEVANT ACTIVITY FOR EACH STUDY IN THE PROJECT

Please see the [Guidelines](#) for the definition of **ethics-relevant** activities.

Study No.	Study Title	Start Date	Duration in Months
1.	Evaluating emotional trust in AI for non-experts.	20/10/2025	6 months (data collect + analysis)
2.	<i>50 words max.</i>		

Please expand the rows in case of additional studies.

SECTION 4: AIM OF EACH OF THE STUDIES

Study No.	Aim	Study will involve: <i>Please tick all appropriate boxes</i>
1.	<i>The aim of this study is to assess the difference, if any, of emotional trust between users engaging with static AI explanations and users engaging with dynamic and visual AI explanations.</i>	☒ Human participants ☐ Vulnerable population ² ☐ Human biological material ☐ Pre-collected biological material ☐ Personal data ☐ Pre-collected personal data ☐ risks on the environment ☐ risks on the society

² Vulnerable populations include pregnant women, neonates, children and adolescents, individuals with cognitive impairment and/or mental disorders, people with anxieties, mentally-deficient persons, members of social minorities, the elderly, and detained persons. *Please consult also page 36 of the Research Ethics Guidelines.*

2.	<i>100 words max.</i>	☐ Human participants ☐ Vulnerable population ☐ Human biological material ☐ Pre-collected biological material ☐ Personal data ☐ Pre-collected personal data ☐ risks on the environment ☐ risks on the society

Please expand the rows in case of additional studies.

SECTION 5: DETAILS OF EACH STUDY

Please specify the ethically relevant issues for each of the studies.

Study 1	Target sample size: (Please justify the target sample size, either statistically or by referring to existing literature)	Literature considers 15-20 participants to be a good start to ensure reliable statistical results. See How Many Participants are Really Enough for Usability Studies?, Roobaea Alroobaea, P.J. Mayhew, August 2014, Conference: Science and Information Conference 2014 August 27-29, 2014 London, UK IEEE, DOI: 10.1109/SAI.2014.6918171
	Key inclusion criteria:	Level of education, limited or low knowledge of AI Visitors of the exhibition that will take place at the Luxembourg Learning Centre, during the Open Data Week (October 20-24).
	Key exclusion criteria:	Inability to provide consent, high knowledge of AI
	Age range of participants:	Legal adults over 18 years old
	Country/ies of recruitment:	Europe
	Expected duration of procedure ³ :	10min
	Data Collection:	<i>Text survey</i>
	Methodology:	<i>The study will consist of a survey of two participants groups. The first group, assigned to an AI visualization, will be the test group, while the group assigned to an AI text explanation will be the Control Group. Participants from the Control Group will be asked to fill in a short survey assessing their general and emotional trust regarding AI. They will then be presented with a detailed text explanation of the AI process from input to output. This explanation will be presented on a single page and will use clear, understandable language to describe the input data, the internal workings and architecture of the neural network model, and the resulting text and image outputs. The content will be conceptually identical to the information conveyed by the visualization, but presented purely as text. Upon finishing reading the text, they will be asked the same questions as before to assess if the textual explanation enhanced, hindered, or did not affect their trust in AI (including emotional trust). Participants from the Test Group will be asked to fill in a short survey assessing their general and emotional trust regarding AI. They will then be confronted with an artistic visualization of the AI process from input to output. This visualization consists of a three screens projection where the first screen refers to the input, represented by a flow of data ; the second screen the inner processing's of the model, represented by visual representations of neural</i>

³ Please indicate the duration the data collection procedure (e.g. 30 min), not of the whole project

		networks ; and the third screen representing the output of the model, that is clear and understandable text and images such as those generated by ChatGPT. Following this, they will be asked the same questions as before to assess if the textual explanation had any change on their perceived trust and understanding of AI. The purpose will be to see whether the artistic depiction and explanation enhanced, hindered, or did not affect their trust in AI (including emotional trust). The project will involve assessment of participants through a general trust scale (TXAI) and an emotional trust scale in AI. The questionnaire is planned to be used during the Open Data Week at the Luxembourg Learning Center (October 20-24). Statistics will be performed on the collected data to determine if there is a difference in emotional trust between participants exposed to visual, interactive and artistic explanations and participants exposed to static text-based explanations. Since this experiment is scheduled during a public event at the Luxembourgish Learning Center, communication staff may record audiovisual content for promotional purposes, that is, film or photography. Participants may appear in this audiovisual content. References Perrig, Sebastian AC, Nicolas Scharowski, and Florian Brühlmann. "Trust issues with trust scales: examining the psychometric quality of trust measures in the context of AI." Extended abstracts of the 2023 CHI Conference on human factors in computing systems. 2023. https://doi.org/10.1145/3544549.3585808 Shang, Ruoxi, Gary Hsieh, and Chirag Shah. "Trusting your ai agent emotionally and cognitively: Development and validation of a semantic differential scale for ai trust." Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society. Vol. 7. 2024. https://doi.org/10.1609/aies.v7i1.31728
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Please expand the table in case of additional studies.

SECTION 6: RISK ASSESSMENT AND RISK MINIMIZATION

A risk assessment should be carried out for each study and it should be demonstrated that appropriate measures are taken to minimize any reasonably foreseeable risk, discomfort or disadvantages.

6.1- Health and safety risk(s) for the HUMAN PARTICIPANTS involved

	Potential risk(s)	Extent of risk(s) and measures taken
<u>Study 1</u>	Human participants are involved in the study with potential risk related to ☐ Physical harm ☐ Psychological harm ☐ Invasion of privacy ☐ Social and economic harm ☐ Other	N/A
	Are there any specific risks to the human participants involved, if any.	<i>Unexpected findings are not considered a risk in this experiment. If the result shows no difference between pre and post exposure to the visualisation, it will be an unexpected finding, however, this poses no risks, as it will not cause any harm to participants nor will it cause harm to the researcher.</i>

Please expand the table in case of additional studies.

6.2- Health and safety risk(s) for the RESEARCHER(S)

	Potential risk(s)	Extent of risk(s) and measures taken
<u>Study 1</u>	Where does the research take place? ☒ Premises of UL ☐ Public area ☐ Study participant's home ☒ Other location <i>The other location will take place online, through the platform Prolific, where we want to repeat the experiment and see if this change of environment affects the results.</i>	N/A
	Are there any specific risks to the researchers (which are not covered by the location)?	<i>Unexpected findings are not considered a risk in this experiment. If the result shows no difference between pre and post exposure to the visualisation, it will be an unexpected finding, however, this poses no risks, as it will not cause any harm to participants nor will it cause harm to the researcher.</i>

Please expand the table in case of additional studies.

6.3- Changes to the ENVIRONMENT and SOCIETY

	Potential risk(s)	Extent of risk(s) and measures taken
Study 1	Research activity includes ☐ Pollutants ☐ Chemical or biological waste ☐ Dual use research of concern (e.g. military research purpose(s)) ☐ Other	N/A
	Is there a risk Have you addressed how unexpected findings will be handled? Are there any specific risks to the environment and/or soc's	

Please expand the table in case of additional studies.

SECTION 7: PARTICIPANTS' INFORMATION AND CONSENT

Participants (or their legal representatives) have to be provided with information on the study, and participants have to give their consent.

Consent form template and instructions can be found on the [University internet site](#).

Please indicate which information and consent form(s) you will use for your study/studies and attach a version of all information (in the language(s) used in the study) given to the participants (or their legal representatives) to this application:

☒ Participants' **information sheet(s)** for study/studies

Please tick as appropriate: ☒ Human participants
☐ Vulnerable population

☒ Participants' **consent** for study/studies

Please tick as appropriate: ☒ Human participants
☐ Vulnerable population

- | | | |
|---|---|---|
| Have you planned an additional oral debriefing ? | ☒ Yes | ☐ No |
| Have you made explicitly clear that participants voluntarily take part in the study? | ☒ Yes | ☐ No |
| Have you made clear that participants can withdraw from the study at any time without consequences? | ☒ Yes | ☐ No |
| Do you plan to do audio/video recording ? | ☒ Yes | ☐ No |
| If you do audio/video recording , did you include this in the informed consent? | ☒ Yes | ☐ No |
| Have you informed participants on the processing and management of personal data in the consent form? | ☒ Yes | ☐ No |
| Do you have the intention to use the data for further research? | ☐ Yes | ☒ No |
| Do you use the informed consent for a single study purpose? | ☒ Yes | ☐ No |
| Have you addressed how unexpected findings will be handled? | ☐ Yes | ☐ No |
| <i>Unexpected findings are those which may reveal situations that suggest the presence of a problem. If you have such eventuality, specify how you intend to handle the situation</i> | | ☒ N/A |
| Have you mentioned to provide the study participant with a copy of the consent? | ☒ Yes | ☐ No |

SECTION 8: DATA PROTECTION

8.1- Anonymity / Pseudonymity

Please describe for each of the studies, how anonymity or pseudonymity is addressed⁴.

*Direct identifiers such as name, email address, telephone or physical address will not be collected. Participants will only be identifiable via tokens, which ensures the anonymity of their data. These tokens, essentially numbers that will be attributed to each respondent, will allow the processing of their data in such a manner that the personal data can no longer be attributed to a specific data subject.
The consent form will be done through Microsoft Forms, so there will be no need for signature, only an "I agree" statement.*

I am aware of and I comply with the University's data protection regulations.



https://www.en.uni.lu/university/data_protection/data_protection_policy_and_faq.

Yes

No

Article 38, related to the position of the DPO of the GDPR and the UL Data Protection Regulation set out the data protection policy of the University.

Further details can be found under:

<https://uniluxembourg.sharepoint.com/sites/dpo/SitePages/Data-Protection-Office.aspx>

In addition to the GDPR regulations, the ERP needs to be informed about pseudonymity – anonymity, data storage and data access (see the following sections).

⁴ Pseudonymization is the processing of personal data in such a manner that the personal data can no longer be attributed to a specific data subject.

Data anonymization is the process of irreversible transformation of personal data.

8.2- Data storage

Please fill in the information in the following table on provisions for integrity and confidentiality.

	<i>Raw data (incl. video or audio recordings)</i>	<i>Processed data (anonymized, pseudonymized)</i>	<i>Document(s) containing the names, contact information and personal details (the pseudonymisation key list)</i>	<i>Audiovi sual material</i>
Data carrier (digitalized or paper form)	<i>Digitalized on Microsoft Forms</i>	<i>Digitalized on Microsoft Forms</i>	<i>Digitalized on Microsoft Forms</i>	<i>Luxemb ourg Learnin g Centre</i>
Number of copies	<i>1 copy with researcher, 1 copy with personal investigator</i>	<i>1 copy on researcher and personal investigator UL laptop, 1 copy on UL server</i>	<i>1 copy with researcher, 1 copy with personal investigator</i>	<i>1 copy with LLC person in charge</i>
Storage Location	<i>UL server, UL laptop</i>	<i>UL server, UL laptop</i>	<i>UL server, UL laptop</i>	<i>LLC server</i>
Retention period	<i>5 years</i>	<i>5 years</i>	<i>5 years</i>	<i>10 years</i>

8.3- Data access

How is access being controlled (physical access, security)?

- *Please list the people who have access to the different types of data under 8.2 – Data storage.*
- *Jules wax (Researcher) and Jacques Klein (Principal Investigator)*
- *Double encryption cryptography is used for managing data access*

ADDITIONAL DOCUMENTS:

The following document(s) has/ have been attached:

- ☐ Previous ethics approvals for the study/studies (a copy of the approval letter, a description of any modifications made to the approved research design)
- ☐ A version of all material used to recruit participants (e.g. flyers, letters).
- ☒ A brief description of equipment used, if relevant.
- ☐ Information sheet(s) for participants, legal guardians and/or other involved parties
- ☒ Consent form(s) for participants, legal guardians and other involved parties
- ☒ A version of all study materials given to the participants (e.g. questionnaires, assignments)
- ☒ A copy of any other information given to the participants
- ☐ Any additional document(s), please specify

Please provide the documents in all languages used in the study.

DECLARATION:

We confirm to conduct the research in an ethically appropriate way in accordance with the University of Luxembourg [Policy on Ethics in Research](#).

Researcher	Name:	Jules Wax
	Signature: (hand-signed)	
	Date:	16/10/2025
Principal Investigator	Name:	Jacques Klein
	Signature: (hand-signed)	
	Date:	16/10/2025

Please expand the rows in case of additional researcher(s) and/or PI(s).